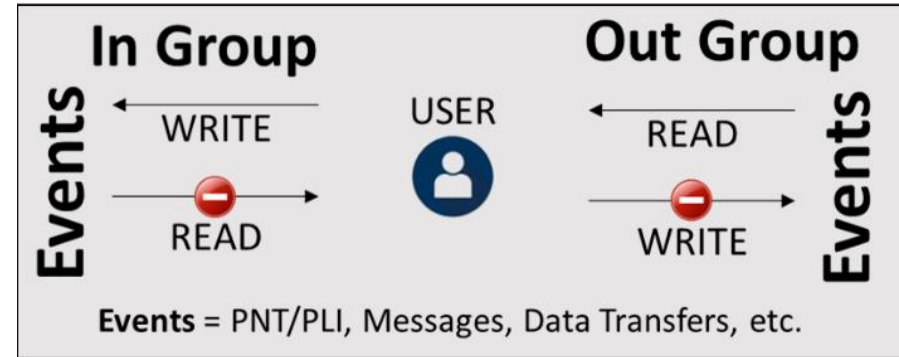


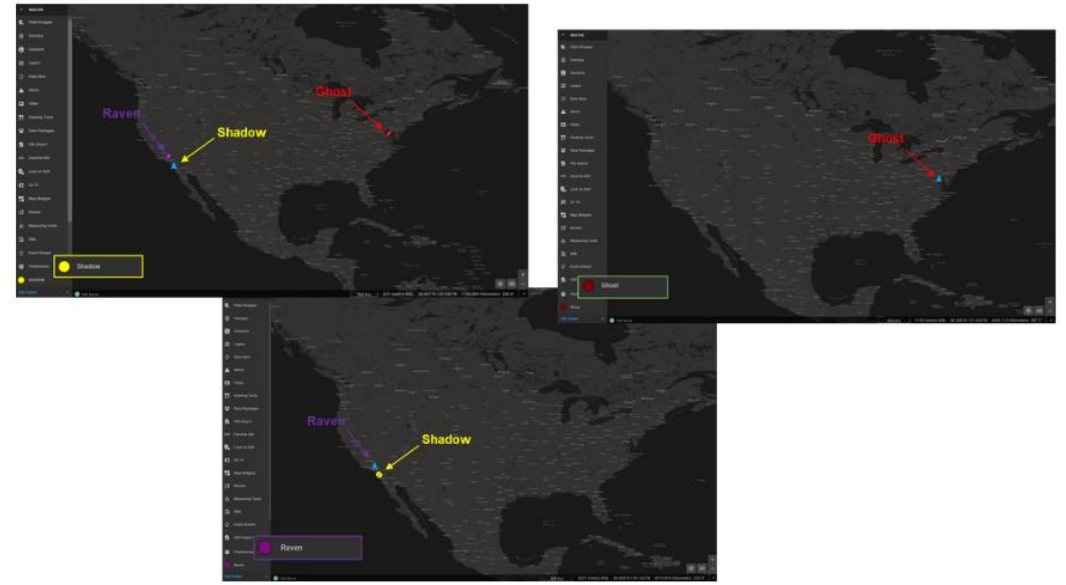
TAK Server - Managing Users and Groups

TAK Server - Groups



- There are three roles within groups:
 - In: the user events are transmitted to the group that they are placed in, allowing the user to write to the server but not read.
 - Out: the user can receive transmitted events from the group they are placed in, allowing the user to read messages but not write.
 - Both: Allows both reading and writing to the server.

▼ Callsign	Username	DN	Groups
SHADOW	Shadow		GroupA←,GroupA→,GroupB→
Raven	Raven		GroupA←,GroupA→
Ghost	Ghost		GroupB←,GroupB→



TAK Server - Implementing Groups - Command Line

- Change to the tak user
 - **sudo su tak**
- Navigate to the certificate directory
 - **cd /opt/tak/certs**
- Create the client certificate
 - **./makeCert.sh client <username>**
 - **Example:** ./makeCert.sh client Denji

```
[takadmin@localhost ~]$ sudo su tak
[sudo] password for takadmin:
bash-4.4$ cd /opt/tak/certs
bash-4.4$ ./makeCert.sh client Denji
client
Denji
Making a client cert for /C=XX/ST=XX/L=XX/O=TEST/OU=TEST/CN=Denji
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'Denji.key'
-----
Signature ok
subject=C = XX, ST = XX, L = XX, O = TEST, OU = TEST, CN = Denji
Getting CA Private Key
Importing keystore Denji.p12 to Denji.jks...
bash-4.4$
```

TAK Server -UserManager Usage Instructions:

- **java -jar UserManager.jar certmod [OPTIONS] certpath**
 - Facilitates the viewing, addition, or modification of the users' using certificates.
- Options:
 - **-A**, --administrator Provides the user with administrative access
 - **-D**, --delete-user Deletes the user. All other optional parameters will be ignored.
 - **-f**, --fingerprint FINGERPRINT Overrides the certificate fingerprint
 - **-p**, --password PASSWORD The password for the user. Password requirements: minimum of 15 characters to include 1 uppercase, 1 lowercase, 1 number, and 1 special character from this list [-_!@#\$%^&*(){}[]+=~`|:;<>,.\/\?]. NOTE: you may need to surround the password string with single quotation marks

TAK Server -UserManager Usage Instructions:

- **java -jar UserManager.jar usermod [OPTIONS] username**
 - Facilitates the viewing, addition, or modification of the users using their username.
- Options:
 - **-A**, --administrator Provides the user with administrative access
 - **-c**, --certificate CERTPATH The filepath to the certificate that contains the user's fingerprint. If the fingerprint is specified as a command line option, this will be ignored!
 - **-D**, --delete-user Deletes the user. All other optional parameters will be ignored.
 - **-f**, --fingerprint FINGERPRINT The fingerprint the user can use to authenticate with
 - **-p**, --password PASSWORD The password for the user. Password requirements: minimum of 15 characters to include 1 uppercase, 1 lowercase, 1 number, and 1 special character from this list [-_!@#\$%^&*(){}[]+=~`|:;<>.,/\?]. NOTE: you may need to surround the password string with single quotation marks

TAK Server -UserManager Usage Instructions:

- To add a user to a specific group:
 - Both Group:
 - **UserManager.jar certmod -g <groupName> /opt/tak/certs/files/<commonName>.pem**
 - **Example:** UserManager.jar certmod -g __TAK__ /opt/tak/certs/files/Denji.pem
 - In Group:
 - **UserManager.jar certmod -ig <groupName> /opt/tak/certs/files/<commonName>.pem**
 - **Example:** UserManager.jar certmod -ig TAK /opt/tak/certs/files/Denji.pem
 - Out Group:
 - **UserManager.jar certmod -og <groupName> /opt/tak/certs/files/<commonName>.pem**
 - **Example:** UserManager.jar certmod -og TEST /opt/tak/certs/files/Denji.pem
- To run the script: (While in the tak user)
 - **java -jar /opt/tak/utils/UserManager.jar certmod -g TEST /opt/tak/certs/files/Denji.pem**
- Note: In the examples different group names were used, if the group doesn't exist, doing this will create the group.

TAK Server - Implementing Groups - Command Line

- Once you are done adding the user to a group you will need to copy over the certificate p12 file if not using auto enrollment.
- You are able to move users from groups if you have already issued the certificate by running the previous commands.
 - With this you will not need to re-create a new cert every time you move, add, or change a user group.

TAK Server - Implementing Groups - Command Line

- Removing Users from Groups:
 - Open and examine the UserAuthentication.xml file:
 - (As the tak user) **cd /opt/tak**
 - **vi UserAuthenticationFile.xml**
 - You are looking for the user you wish to remove and their group you wish to remove them from, which will be between:
 - Both group: `<groupList> <group> </group> </groupList>`
 - In Group: `<groupListIN> <group> </groupListIN>`
 - Out Group: `<groupListOUT> <group> </groupListOUT>`
 - Once you find the group of the user you want to remove them from exit out of the file without editing it:
 - Press “esc”
 - **:q!**
 - Press enter

```
<groupList>__ANON__</groupList>
</User>
<User identifier="Denji" fingerprint="3A:14:37:62:98:DB:8E:21:AD:A9:C2:27:2F:82:58:8E:35:A5:63:28:2C:77:64:3F:31:
1D:6C:54:5E:D4:65:31" passwordHashed="false">
  <groupList>TEST</groupList>
```


TAK Server - Implementing Groups - Command Line

- As the tak user
 - Remove the user from GroupB but retain GroupA
 - **java -jar /opt/tak/utils/UserManager.jar certmod -g GroupA /opt/tak/certs/files/<user>.pem**
 - Remove the user from GroupA but retain Out Group in GroupB
 - **java -jar /opt/tak/utils/UserManager.jar certmod -og GroupB /opt/tak/certs/files/<user>.pem**
 - To remove a user from a group you only specify the groups you want them to be in and omit the groups you don't.
 - You can add them to multiple groups by adding them next to each other as follows:
 - **java -jar /opt/tak/utils/UserManager.jar certmod -g GroupA -og GroupB -ig GroupC /opt/tak/certs/files/<user>.pem**

```
bash-4.4$ java -jar /opt/tak/utils/UserManager.jar certmod -g TEST -og __ANON__ -ig TAK /opt/tak/certs/files/Denji.pem
User Updated:
Username:      'Denji'
Role:          ROLE_ANONYMOUS
Fingerprint:   3A:14:37:62:98:DB:8E:21:AD:A9:C2:27:2F:82:58:8E:35:A5:63:28:2C:77:64:3F:31:1D:6C:54:5E:D4:65:31
Groups (read and write permission):
    TEST
Groups IN (write permission):
    TAK
Groups OUT (read permission):
    __ANON__
bash-4.4$ _
```

TAK Server - Implementing Groups - Server Dashboard

- Open the server dashboard with a web browser and navigate to the Manager Users section:
 - Top right hamburger menu → Administrative → Manage Users
- From here you are able to:
 - Add users one at a time: Top left, select “Add User” .
 - Add users in bulk: Top left, select “Add Users”.
 - Manage groups:
 - Add new group: Top right, select “Add New Group”.
 - Search for a group.
 - View the users in a group: select the group → List users

TAK Server - Implementing Groups - Server Dashboard

- You can add a new user or select a current user.
 - New user:
 - Provide a username and password for them.
 - On the right drag and drop the desired groups where you'd like them.
 - When you are done, select “Create New User”
 - Existing user:
 - select the user from the “Users” panel and then select “View/Edit groups”.
 - You can drag and drop the groups currently within the user or from the right hand “Groups” panel to the desired locations.
 - When you are done, select “Update”

The screenshot shows the 'Add New User' form. It has a 'Username' field with the value 'TEST' and a 'Password' field with masked characters. To the right of the password field are 'Create New User' and 'Cancel' buttons. Below the fields is a set of three boxes labeled 'In Group', 'Out Group', and 'Both'. The 'In Group' box contains a 'TAK' button. The 'Out Group' box contains a 'TEST' button. The 'Both' box contains 'TEST' and 'ANON' buttons. To the right of these boxes is a 'Groups' panel with a search bar and a list of groups: 'ANON', 'TEST', 'TAK', and 'TEST'. Each group has a dropdown arrow. A note below the boxes says: 'Drag the groups on the right panel and drop to the boxes below. To remove a group from the boxes, drag it back to the right panel. If no groups are specified, user will be in group '_ANON_'.'

The screenshot shows the 'User-Group Management' form. It has a title 'User-Group Management' and a subtitle 'Edit groups for user test'. There are 'Update' and 'Reset' buttons. Below the buttons is a set of three boxes labeled 'In Group', 'Out Group', and 'Both'. The 'In Group' box contains an 'ANON' button. The 'Out Group' box contains a 'TEST' button. The 'Both' box contains 'TEST' and 'TAK' buttons. To the right of these boxes is a 'Groups' panel with a search bar and a list of groups: 'ANON', 'TEST', 'TAK', and 'TEST'. Each group has a dropdown arrow. A note below the boxes says: 'Drag the groups on the right panel and drop to the boxes below. (To remove a group from the boxes, drag it back to the right panel)'

TAK Server - Using Input Based Grouping

- Determine Open Ports:
 - View active listening ports (based on your distro some commands will not be available)
 - **sudo lsof -i -P -n | grep LISTEN**
 - **sudo netstat -tulpn | grep LISTEN**
 - **sudo ss -tulpn | grep LISTEN**
 - You only need to run one of the above command, which ever works.
 - We want to identify that the port we have in mind is not already in use.
 - Example: 8087

```
systemd      1      root    30u    IPv4    20610      0t0  TCP *:111 (LISTEN)
systemd      1      root    33u    IPv6    20612      0t0  TCP *:111 (LISTEN)
rpcbind      775     rpc      4u     IPv4    20610      0t0  TCP *:111 (LISTEN)
rpcbind      775     rpc      6u     IPv6    20612      0t0  TCP *:111 (LISTEN)
sshd         979     root     3u     IPv4    28289      0t0  TCP *:22 (LISTEN)
sshd         979     root     4u     IPv6    28299      0t0  TCP *:22 (LISTEN)
cupsd        980     root     6u     IPv6    26016      0t0  TCP [::]:631 (LISTEN)
cupsd        980     root     7u     IPv4    26017      0t0  TCP 127.0.0.1:631 (LISTEN)
postmaster   1321    postgres 6u     IPv6    32296      0t0  TCP [::]:5432 (LISTEN)
postmaster   1321    postgres 7u     IPv4    32297      0t0  TCP 127.0.0.1:5432 (LISTEN)
dnsmasq      1980    dnsmasq  6u     IPv4    36229      0t0  TCP 192.168.122.1:53 (LISTEN)
java         12998   tak      14u    IPv4    82474      0t0  TCP 127.0.0.1:47104 (LISTEN)
java         12998   tak      25u    IPv4    83263      0t0  TCP 127.0.0.1:10804 (LISTEN)
java         13056   tak      11u    IPv4    79492      0t0  TCP 127.0.0.1:47100 (LISTEN)
java         13056   tak      20u    IPv4    79501      0t0  TCP 127.0.0.1:47500 (LISTEN)
java         13056   tak      23u    IPv4    82195      0t0  TCP 127.0.0.1:10800 (LISTEN)
java         13056   tak      34u    IPv4    82217      0t0  TCP 127.0.0.1:11211 (LISTEN)
java         13056   tak      93u    IPv4    83308      0t0  TCP *:8089 (LISTEN)
java         13101   tak      11u    IPv4    83047      0t0  TCP 127.0.0.1:47102 (LISTEN)
java         13101   tak      22u    IPv4    83155      0t0  TCP 127.0.0.1:10802 (LISTEN)
java         13101   tak      42u    IPv4    82507      0t0  TCP *:8446 (LISTEN)
java         13101   tak      54u    IPv4    83321      0t0  TCP *:8443 (LISTEN)
java         13101   tak      58u    IPv4    82538      0t0  TCP *:8444 (LISTEN)
java         13152   tak      12u    IPv4    79544      0t0  TCP 127.0.0.1:47101 (LISTEN)
java         13152   tak      23u    IPv4    82411      0t0  TCP 127.0.0.1:10801 (LISTEN)
java         13208   tak      12u    IPv4    79688      0t0  TCP 127.0.0.1:47103 (LISTEN)
java         13208   tak      23u    IPv4    83187      0t0  TCP 127.0.0.1:10803 (LISTEN)
[takadmin@localhost tak]$
```

TAK Server - Using Input Based Grouping

- Add the port to our firewall:
 - **sudo firewall-cmd --zone=public --add-port 8087/tcp --permanent**
- Reload the firewall to accept the changes:
 - **sudo firewall-cmd --reload**
- Verify the changes:
 - **sudo firewall-cmd --list-all**

```
[takadmin@localhost tak]$ sudo firewall-cmd --zone=public --add-port 8087/tcp --permanent
success
[takadmin@localhost tak]$ sudo firewall-cmd --reload
success
[takadmin@localhost tak]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3 enp0s8
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 8089/tcp 8443/tcp 8446/tcp 8087/tcp
  protocols:
  forward: no
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[takadmin@localhost tak]$
```

TAK Server - Using Input Based Grouping

- Open the admin dashboard → hamburger menu → Configuration → Input and Data Feeds.
- Select “Add Input (Server Port)” :
 - Name: (Your Choice) → Testing
 - Protocol: Standard Streaming TCP (TLS)
 - Authentication Type: X.509
 - Port: (The port you configured) → 8087
 - Filter Groups: GroupA
 - These can be new groups or existing groups from other ports.
 - You are able to add multiple groups but add each to their own line.
 - Select “Save”

Create Input Definition

*Name
Name may contain upper and lower case letters, numbers, spaces and underscores up to 30 characters.

*Protocol

*Authentication Type

*Port
Port must be an integer between 1 and 65535.

Core Messaging Version

Core Messaging 2 TLS Version(s):
TLSv1: ☐ TLSv1.1: ☐ TLSv1.2: ☒ TLSv1.3: ☒

Multicast Group
Provide an optional valid IPv4 multicast address.

Interface
Provide an optional valid network interface name (e.g., en0).

Archive

Anonymous Group

Archive Only

Filter Groups
Provide 0 or more LDAP distinguished names separated by newlines.
Example: cn=John Smith,ou=test,dc=xyz,dc=gov

*Required

TAK Server - Using Input Based Grouping - Command Line

- Adding inputs editing the Core Configuration file:
 - Navigate to and open the CoreConfig.xml file
 - **cd /opt/tak**
 - **vi CoreConfig.xml**
 - In-between our <network> </network> section add the following:
 - <input auth="x509" _name="Testing" protocol="tls" port="8087" archive="true" anongroup="false" archiveOnly="false" coreVersion="2" coreVersion2TlsVersions="TLSv1.2,TLSv1.3">
 - <filtergroup>GroupA</filtergroup>
 - <filtergroup>GroupD</filtergroup>
 - </input>
 - You can change the values as you desire to add different groups or change the port type.

```
<network multicastTTL="5" serverId="d719bdaee6d9499da469a49b91655132" version="5.2-RELEASE-16-HEAD">
  <input _name="stdssl" protocol="tls" port="8089" coreVersion="2"/>
  <input auth="x509" _name="Testing" protocol="tls" port="8087" archive="true" anongroup="false" archiveOnly="false" coreVersion="2" coreVersion2TlsVersions="TLSv1.2,TLSv1.3">
    <filtergroup>GroupA</filtergroup>
    <filtergroup>GroupB</filtergroup>
  </input>
  <connector port="8443" _name="https"/>
  <connector port="8444" useFederationTruststore="true" _name="fed_https"/>
  <connector port="8446" clientAuth="false" _name="cert_https"/>
  <announce/>
</network>
```

TAK Server - Input Based Grouping - Connection

- When attempting to connect a TAK device you will follow the standard process but update the port option to that of the configured one.
- Once connected you can verify the settings and number of connected devices from the “Inputs and Data Feeds” panel.

Description

test

Protocol

Host Address

Port

SSL

192.168.1.126

8087

Certificates

☒ Enroll for Client Certificate

Install Certificate Authority

Install Client Certificate

☐ Use Authentication

Username:

Password:

OK

Cancel

Inputs (Server Ports)

[Add Input \(Server Port\)](#)

Auth Type	Name	Protocol	Port	Core Messaging Version	TLS	Group	Interface	Archive	Filter Group	Anonymous Group	Archive Only	Reads Received	Messages Received	Number Connected	Bytes Receive
X_509	Testing	tls	8087	2	TLSv1.2,TLSv1.3			true	["GroupA","TEST"]	false	false	143	134	2	46616
X_509	stdssl	tls	8089	2	TLSv1.2,TLSv1.3			true	[]		false	0	0	0	0